

Complex Number

↳ An ordered pair of real numbers (a, b) in which 'a' is called the real part and 'b' is called the imaginary part is known as complex number.

↳ Complex number (a, b) can be written as $a + ib$.

↳ $i = \sqrt{-1}$

Algebra of complex numbers

1. Equality

↳ Two complex numbers $Z = a + ib$ and $W = c + id$ are equal if and only if $a = c$ & $b = d$.

2. Addition

↳ $(a, b) + (c, d) = (a + c, b + d)$

3. Subtraction

↳ $(a, b) - (c, d) = (a - c, b - d)$

4. Multiplication

↳ $(a, b) \cdot (c, d) = (ac - bd, ad + bc)$

5. Division

↳ $\frac{(a, b)}{(c, d)} = \left(\frac{ac + bd}{c^2 + d^2}, \frac{bc - ad}{c^2 + d^2} \right); (c, d) \neq 0$

Note: The set of all complex numbers is denoted by $\mathbb{C}$.

So,

$$\mathbb{C} = \{z: z = a + ib; a, b \in \mathbb{R}\}$$

• The set of real numbers $\mathbb{R}$ is a proper subset of set of complex number $\mathbb{C}$.
i.e. $\mathbb{R} \subset \mathbb{C}$

• $\sqrt{a} \times \sqrt{b} = \sqrt{ab}$ holds only if at least one of a or b is +ve

The imaginary unit

↳ The complex number $(0, 1)$ denoted by 'i' is called imaginary unit.

Note that: $i^2 = -1$

Proof

$$\begin{aligned} i^2 &= i \cdot i \\ &= (0, 1) \cdot (0, 1) \\ &= (0 - 1, 0 + 0) \\ &= (-1, 0) \\ &= (-1) \end{aligned}$$

$$\therefore i^2 = -1$$

Proved

* Integral Powers of i

$$\bullet i^1 = i$$

$$\bullet i^4 = 1$$

$$\bullet i^2 = -1$$

$$\bullet i^3 = i^2 \cdot i = -i$$

Note: An Integral power of i is one of the four numbers $i, -1, -i$ or 1 and repeats in a cycle.

• For any integer n ,

$$i^{4n} = (i^4)^n = 1^n = 1$$

Exercise 7.1

1. Write the following complex numbers in the cartesian form i.e. $a+ib$:

a) $(3, 4)$

↳ $3+4i$

b) $(0, 2)$

↳ $0+2i$

c) $(-5, 0)$

↳ $-5+0i$

d) $(2, -6)$

↳ $2-6i$

2. Write the following complex numbers in the form of ordered pair i.e. (a, b) :

a) $5-6i$

↳ $(5, -6)$

b) $3i$

↳ $(0, 3)$

c) 7

↳ $(7, 0)$

d) $8+2i$

↳ $(8, 2)$

3. Find real and imaginary parts of the following complex numbers:

a) $(2, -5)$

↳ Real part $\div 2$

Imaginary part $\div -5$

b) $(0, 3)$

↳ Real part $\div 0$

Imaginary part $\div 3$

c) $(-6, 0)$

↳ Real part $\div -6$

Imaginary part $\div 0$

d) $1 + 4i$

↳ Real part $\div 1$

Imaginary part $\div 4$

e) $-3i$

↳ Real part $\div 0$

Imaginary part $\div -3$

f) $\frac{1}{5}$

↳ Real part $\div \frac{1}{5}$

Imaginary part $\div 0$

g) $\frac{1}{5}i$

$= \frac{i^4}{-5i} \quad [i^4 = 1]$

$= \frac{i^3}{-5}$

$= \frac{i^3}{-5}$

$= \frac{-i}{5}$

$= \frac{-i}{5}$

↳ Real part $\div 0$

Imaginary part $\div \frac{-1}{5}$

h) $\frac{1}{3+4i}$

$= \frac{3-4i}{3^2 - (4i)^2}$

$= \frac{3-4i}{9-16i^2}$

$= \frac{3-4i}{9+16}$

$= \frac{3-4i}{25}$

$= \frac{3-4i}{25}$

$= \frac{3-4i}{25}$

$= \frac{3-4i}{25}$

$= \frac{3-4i}{25}$

$= \frac{3-4i}{25}$

↳ Real part $\div \frac{3}{25}$

Imaginary part $\div \frac{-4}{25}$

$$\begin{aligned} i) & (2+3i)(5-4i) \\ &= 2(5-4i) + 3i(5-4i) \\ &= 10 - 8i + 15i - 12i^2 \\ &= 10 + 7i + 12 \\ &= 7i + 22 \\ &= 22 + 7i \end{aligned}$$

↳ Real part $\div 22$
Imaginary part $\div 7$

$$\begin{aligned} j) & (2, -1) \\ & (3, 2) \\ &= \frac{2-i}{3+2i} \\ &= \frac{(2-i)(3-2i)}{9-4i^2} \\ &= \frac{6-3i-4i+2i^2}{9+4} \\ &= \frac{6-7i-2}{13} \\ &= \frac{4}{13} - \frac{7i}{13} \end{aligned}$$

↳ Real part $\div \frac{4}{13}$

Imaginary part $\div -\frac{7}{13}$

4. Evaluate

$$\begin{aligned} a) & (1, 0)^{15} \\ &= (1+0i)^{15} \\ &= 1^{15} \\ &= 1 \end{aligned}$$

$$\begin{aligned} b) & (1, 0)^{-24} \\ &= (1+0i)^{-24} \\ &= 1^{-24} \\ &= 1 \end{aligned}$$

$$\begin{aligned} c) & (0, 1)^8 \\ &= (0+i)^8 \\ &= i^8 \\ &= (i^4)^2 \\ &= 1^2 \\ &= 1 \end{aligned}$$

$$\begin{aligned} d) & (0, 1)^{-37} \\ &= (0+i)^{-37} \\ &= i^{-37} \\ &= \frac{1}{i^{37}} \\ &= \frac{i^{40}}{i^{37}} \quad [\because (i^4)^{10} = 1^{10} = 1] \\ &= i^3 = -i \end{aligned}$$

e. i^{48}
 $= (i^4)^{12}$
 $= (1)^{12}$
 $= 1$

f) i^{-97}
 $= \frac{1}{i^{97}}$
 $= \frac{i^{100}}{i^{97}} \quad [\because i^{100} = (i^4)^{25} = 1]$
 $= i^3$
 $= -i$

5. Find $Z_1 + Z_2$, $Z_1 - Z_2$, $Z_1 \cdot Z_2$ and $\frac{Z_1}{Z_2}$ if

a) $Z_1 = (2, -3)$ and $Z_2 = (-3, -4)$

Soln

$$\begin{aligned} Z_1 + Z_2 &= (2, -3) + (-3, -4) \\ &= (2-3, -3-4) \\ &= (-1, -7) \end{aligned}$$

$$\begin{aligned} Z_1 - Z_2 &= (2, -3) - (-3, -4) \\ &= (2 - (-3), -3 - (-4)) \\ &= (5, 1) \end{aligned}$$

$$\begin{aligned} Z_1 \cdot Z_2 &= (2, -3) \cdot (-3, -4) \\ &= (2-3i)(-3-4i) \\ &= -6 - 8i + 9i + 12i^2 \\ &= -6 + i - 12 \\ &= -18 + i \\ &= (-18, 1) \end{aligned}$$

$$\begin{aligned} \frac{Z_1}{Z_2} &= \frac{(2, -3)}{(-3, -4)} \\ &= \frac{2-3i}{-3-4i} \end{aligned}$$

$$\begin{aligned}
 &= \frac{2-3i}{-(3+4i)} \\
 &= \frac{(-2+3i) \times (3-4i)}{(3+4i)(3-4i)} \\
 &= \frac{-6+8i+9i-12i^2}{9-16i^2} \\
 &= \frac{-6+17i+12}{9+16} \\
 &= \frac{6+17i}{25} \\
 &= \frac{6}{25} + \frac{17i}{25} \\
 &= \left(\frac{6}{25}, \frac{17}{25} \right)
 \end{aligned}$$

b) ~~$Z_1 = 1-6i$~~ and $Z_2 = 8-6i$

Soln.

$$\begin{aligned}
 Z_1 + Z_2 &= 1-6i + 8-6i \\
 &= 9-12i
 \end{aligned}$$

$$\begin{aligned}
 Z_1 - Z_2 &= 1-6i - (8-6i) \\
 &= 1-6i-8+6i \\
 &= -7
 \end{aligned}$$

$$\begin{aligned}
 Z_1 \cdot Z_2 &= (1-6i)(8-6i) \\
 &= 8-6i-48i+36i^2 \\
 &= 8-54i-36 \\
 &= -28-54i
 \end{aligned}$$

$$\begin{aligned}
 \frac{Z_1}{Z_2} &= \frac{1-6i}{8-6i} \\
 &= \frac{(1-6i)(8+6i)}{8^2-(6i)^2} \\
 &= \frac{8+6i-48i-36i^2}{64-36i^2} \\
 &= \frac{8-42i+36}{64+36} \\
 &= \frac{44-42i}{100} \\
 &= \frac{2(22-21i)}{2 \times 50} \\
 &= \frac{22}{50} - \frac{21}{50}i \\
 &= \frac{11}{25} - \frac{21}{50}i
 \end{aligned}$$

Properties of addition and multiplication of complex numbers.

→ For any three complex numbers z_1, z_2 and z_3

① Closure property

$$z_1 + z_2 \in \mathbb{C}$$

$$\text{and } z_1 \times z_2 \in \mathbb{C}$$

② Commutative property

$$z_1 + z_2 = z_2 + z_1$$

$$\text{and } z_1 \times z_2 = z_2 \times z_1$$

③ Associative property

$$(z_1 + z_2) + z_3 = z_1 + (z_2 + z_3)$$

$$\text{and } (z_1 \times z_2) \times z_3 = z_1 \times (z_2 \times z_3)$$

④ Distributive property

$$z_1 \times (z_2 + z_3) = z_1 \times z_2 + z_1 \times z_3$$

$$\text{and } (z_1 + z_2) \times z_3 = z_1 \times z_3 + z_2 \times z_3$$

⑤ Existence of identity element

- Additive identity

→ The complex number $0 = 0 + 0i$ is called additive identity because $z + 0 = z = 0 + z$ for all $z \in \mathbb{C}$

- Multiplicative identity

→ The complex number $1 = 1 + 0i$ is called multiplicative identity because $z \times 1 = z = 1 \times z$ for all $z \in \mathbb{C}$

⑥ Existence of inverse

• Additive inverse

→ A complex number $-z$ is called additive inverse of z because $z + (-z) = 0 = (-z) + z$ for all $z \in \mathbb{C}$

• Multiplicative inverse

→ A complex number $\frac{1}{z}$ is called multipl-

icative inverse of $z \neq 0$ because $z \times \frac{1}{z} = 1$

$= \frac{1}{z} \times z$ for all $z \in \mathbb{C} - \{0\}$

Note: The additive inverse of $z = (a, b)$ is $-z = (-a, -b)$.

Exercise

6. Find additive identity element and multiplicative identity element of

a) $5 + 6i$

→ Soln,

Additive identity element of $5 + 6i = (0, 0)$

Multiplicative identity element of $5 + 6i = (1, 0)$

b) $(3, -4)$

→ Soln,

Additive identity element of $(3, -4)$ is $(0, 0)$

Multiplicative identity element of $(3, -4)$ is

(1,0).

7. Find additive and multiplicative inverse of

a) $-3-4i$

→ Additive inverse of $-3-4i = -(-3-4i)$
 $= (3+4i)$

Multiplicative inverse of $-3-4i = \frac{1}{-3-4i}$

$$\begin{aligned} &= \frac{-1}{3+4i} \\ &= \frac{-1}{3+4i} \times \frac{3-4i}{3-4i} \\ &= \frac{-3+4i}{9-16i^2} \\ &= \frac{-3+4i}{9+16} \\ &= \frac{-3+4i}{25} \\ &= \frac{-3}{25} + \frac{4}{25}i \end{aligned}$$

4.

b) $(8,-6)$

→ Additive inverse of $(8,-6) = -(8,-6)$
 $= (-8,6)$

Multiplicative inverse of $(8,-6) = \frac{1}{(8,-6)}$

$$= \frac{1}{8-6i}$$

$$= \frac{1}{8-6i} \times \frac{8+6i}{8+6i}$$

$$= \frac{8+6i}{8^2-6^2i^2}$$

$$= \frac{8+6i}{64-36i^2}$$

$$= \frac{8+6i}{64+36}$$

$$= \frac{8+6i}{100}$$

$$= \frac{82}{100} + \frac{36}{100}i$$

$$= \left(\frac{2}{25}, \frac{3}{50} \right)$$

↓.

$$c) \frac{(1, -2)}{(-3, 4)}$$

$$= \frac{1-2i}{-3+4i}$$

$$= \frac{2i-1}{3-4i} \times \frac{3+4i}{3+4i}$$

$$= \frac{2i(3+4i) - 1(3+4i)}{3^2 - 16i^2}$$

$$= \frac{6i + 8i^2 - 3 - 4i}{9 + 16}$$

$$= \frac{2i - 8 - 3}{25}$$

$$= \frac{2i - 11}{25}$$

$$= -\frac{11}{25} + \frac{2}{25}i$$

Now,

$$\text{Additive inverse of } \frac{(1, -2)}{(1-3, 4)} = -\left(-\frac{11}{25} + \frac{2i}{25}\right) \\ = \left(\frac{11}{25}, -\frac{2}{25}\right)$$

$$\begin{aligned} \text{Multiplicative inverse} &= \frac{1}{-\frac{11}{25} + \frac{2i}{25}} \\ &= \frac{1 \times 25}{-11 + 2i} \\ &= \frac{-25}{11 - 2i} \\ &= \frac{-25(11 + 2i)}{(11 - 2i)(11 + 2i)} \\ &= \frac{-275 - 50i}{121 - 4i^2} \\ &= \frac{-275 - 50i}{121 + 4} \\ &= \frac{-275}{125} - \frac{50i}{125} \\ &= \left(-\frac{11}{5}, -\frac{2}{5}\right) \end{aligned}$$

$$\begin{aligned} \text{d) } &\frac{7 + 3i}{-6 - 8i} \\ &= \frac{7 + 3i}{-(6 + 8i)} \\ &= \frac{-(7 + 3i)}{6 + 8i} \times \frac{6 - 8i}{6 - 8i} \\ &= \frac{-(7 + 3i)(6 - 8i)}{36 - 64i^2} \end{aligned}$$

$$\begin{aligned}
 &= \frac{(7+3i)(8i-6)}{36+64} \\
 &= \frac{56i+24i^2-42-18i}{100} \\
 &= \frac{38i-24-42}{100} \\
 &= \frac{38i-66}{100} \\
 &= \frac{19i-33}{50}
 \end{aligned}$$

Now,

$$\begin{aligned}
 \text{Additive inverse of } \frac{7+3i}{-6-8i} &= -\left(\frac{19i-33}{50}\right) \\
 &= \left(\frac{33}{50} - \frac{19i}{50}\right) \\
 &= \left(\frac{33}{50} - \frac{19i}{50}\right)
 \end{aligned}$$

$$\text{Multiplicative inverse of } \frac{7+3i}{-6-8i} = \frac{1}{\frac{19i-33}{50}}$$

*

$$\begin{aligned}
 &= \frac{50}{19i-33} \\
 &= \frac{-50}{33-19i} \\
 &= \frac{-50(33+19i)}{(33)^2 - (19)^2 i^2} \\
 &= \frac{-1650 - 950i}{1089 - 361i^2} \\
 &= \frac{-1650 - 950i}{1089 + 361}
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{-1650 - 950i}{1450} \\
 &= \frac{50(-33 - 19i)}{29 \times 50} \\
 &= \frac{-33}{29} - \frac{19}{29}i \quad \#
 \end{aligned}$$

8. Find the values of x and y if

a) $(x, y) = (2, -3) + (-4, -6)$

or, $(x, y) = (2 - 4, -3 - 6)$

$(x, y) = (-2, -9)$

$\therefore x = -2$

$y = -9 \quad \#$

b) $(6, -4) = (-2, -1) + (x, y)$

or, $(x, y) = (6, -4) - (-2, -1)$

or, $(x, y) = (6 - (-2), -4 - (-1))$

$(x, y) = (8, -3)$

$\therefore x = 8$

$y = -3 \quad \#$

c) $(x, y) = (2, 3) (-1, -2)$

$= (2 + 3i) (-1 - 2i)$

$= -2 - 4i - 3i - 6i^2$

$= -2 - 7i + 6$

$= 4 - 7i$

$= (4, -7)$

$\therefore x = 4$

$y = -7 \quad \#$

$$\begin{aligned}
 d) (x, y) &= \frac{(-1, 5)}{(-3, 4)} \\
 &= \frac{-1 + 5i}{-3 + 4i} \\
 &= \frac{-1 + 5i}{4i - 3} \times \frac{4i + 3}{4i + 3} \\
 &= \frac{-4i - 3 + 20i^2 + 15i}{16i^2 - 9} \\
 &= \frac{-11i - 3 - 20}{-16 - 9} \\
 &= \frac{-23 + 11i}{-25} \\
 &= \frac{23 - 11i}{25} \\
 &= \left(\frac{23}{25}, \frac{-11}{25} \right) \#.
 \end{aligned}$$

$$\begin{aligned}
 e) 3x - iy &= (1 + 2i)^2 \\
 \text{or, } 3x - iy &= 1 + 2 \cdot 1 \cdot 2i + 4i^2 \\
 \text{or, } 3x - iy &= 1 + 4i + 4i^2 \\
 \text{or, } 3x - iy &= 1 + 4i - 4 \\
 \text{or, } 3x - iy &= -3 + 4i
 \end{aligned}$$

Now,

Comparing L.H.S with R.H.S

$$\begin{aligned}
 3x &= -3 & -iy &= 4i \\
 x &= -1 & y &= -4
 \end{aligned}$$

$$\begin{aligned}
 \therefore x &= -1 \\
 y &= -4 \#.
 \end{aligned}$$

$$f) (x+2) + (3-y)i = (-1+i) \cdot (3+2i)$$

$$\text{or, } (x+2) + (3-y)i = -3 + 3i - 2i + 2i^2$$

$$\text{or, } (x+2) + (3-y)i = -3 + i - 2$$

$$\text{or, } (x+2) + (3-y)i = -5 + i$$

Comparing L.H.S With R.H.S

$$x+2 = -5$$

$$3-y = 1$$

$$x = -5 - 2$$

$$y = 3 - 1$$

$$x = -7$$

$$y = 2$$

$$\therefore x = -7$$

$$y = 2$$

9. Simplify and show that

a) $\frac{5-2i}{4-3i} + \frac{7+3i}{4+3i}$ is a complex number

→ Soln,

$$\begin{aligned} & \frac{5-2i}{4-3i} + \frac{7+3i}{4+3i} \\ &= \frac{(5-2i) \cdot (4+3i) + (7+3i) \cdot (4-3i)}{(4+3i)(4-3i)} \\ &= \frac{20 - 8i + 15i - 6i^2 + 28 + 12i - 21i - 9i^2}{16 - 9i^2} \\ &= \frac{48 - 2i + 6 + 9}{16 + 9} \\ &= \frac{63 - 2i}{25} \\ &= \frac{63}{25} - \frac{2}{25}i \end{aligned}$$

Here, the solution is in the form of a + bi

ib, so, it is complex number.

b) $\frac{6-8i}{6+8i} + \frac{6+8i}{6-8i}$ is purely a real number.

$$= \frac{(6-8i)^2 + (6+8i)^2}{(6+8i)(6-8i)}$$

$$= \frac{36 - 2 \cdot 6 \cdot 8i + 64i^2 + 36 + 2 \cdot 6 \cdot 8i + 64i^2}{36 - 64i^2}$$

$$= \frac{72 - 64 - 64}{36 + 64}$$

$$= \frac{-56}{100}$$

$$= \frac{-14}{25}$$

Here, $\frac{-14}{25}$ is a purely real number because it has no imaginary part.

10. If $z = 2+3i$ and $w = 2-3i$, find the value of

a) $z^2 + w^2$

$$= (2+3i)^2 + (2-3i)^2$$

$$= 4 + 2 \cdot 2 \cdot 3i + 9i^2 + 4 - 2 \cdot 2 \cdot 3i + 9i^2$$

$$= 4 - 9 + 4 - 9$$

$$= 8 - 9 - 9$$

$$= -10$$

∴

b) $z^2 - w^2$

$= (2+3i)^2 - (2-3i)^2$

$= 4 + 2 \cdot 2 \cdot 3i + 9i^2 - (4 - 2 \cdot 2 \cdot 3i + 9i^2)$

$= \cancel{4} + 12i + \cancel{9i^2} - \cancel{4} + 12i - \cancel{9i^2}$

$= 24i$

c) $z^2 - zw + w^2$

$= z^2 + w^2 - zw$

$= -10 - (2+3i)(2-3i) [\because z^2 + w^2 = -10]$

$= -10 - 4 + \cancel{6i} - \cancel{6i} + 9i^2$

$= -14 - 9$

$= -23 \#.$

d) $(z-w)^2$

$= [2+3i - (2-3i)]^2$

$= (\cancel{2}+3i - \cancel{2}+3i)^2$

$= (6i)^2$

$= 36i^2$

$= -36 \#.$

11. Simplify

a) $(3-\sqrt{-16})(2+\sqrt{-9})$

$= (3-\sqrt{-1 \times 4 \times 4})(2+\sqrt{-1 \times 3 \times 3})$

$= (3-4i)(2+3i)$

$= 6 + 9i - 8i - 12i^2$

$= 6 + i + 12$

$= 18 + i$

$$\begin{aligned}
 b) & \sqrt{-25} \cdot \sqrt{-16} \\
 &= \sqrt{-1} \cdot \sqrt{25} \cdot \sqrt{-1} \cdot \sqrt{16} \\
 &= 5i \cdot 4i \\
 &= 20i^2 \\
 &= 20 \times -1 \\
 &= -20
 \end{aligned}$$

$$\begin{aligned}
 c) & \sqrt{-2} \cdot \sqrt{-4} \\
 &= \sqrt{-1} \cdot \sqrt{2} \cdot \sqrt{-1} \cdot \sqrt{4} \\
 &= i\sqrt{2} \cdot i\sqrt{4} \\
 &= i\sqrt{2} \cdot 2i \\
 &= 2\sqrt{2}i^2 \\
 &= -2\sqrt{2} [\because i^2 = -1]_{\#}
 \end{aligned}$$

$$\begin{aligned}
 d) & i^{10} + i^{25} + i^{30} + i^{245} \\
 &= (i^4)^2 \cdot i^2 + (i^4)^6 \cdot i + (i^2)^{15} + (i^4)^{61} \cdot i \\
 &= 1 \cdot -1 + 1 \cdot i + (-1)^{15} + 1 \cdot i \\
 &= -1 + i - 1 + i \\
 &= -2 + 2i
 \end{aligned}$$

$$\begin{aligned}
 e) & (0,1)^5 + (0,1)^{14} - (0,1)^{10} - (0,1)^{17} \\
 &= (0+i)^5 + (0+i)^{14} - (0+i)^{10} - (0+i)^{17} \\
 &= i^5 + i^{14} - i^{10} - i^{17} \\
 &= i^4 \cdot i + (i^2)^7 - (i^2)^5 - (i^4)^2 \cdot i \\
 &= 1 \cdot i + (-1)^7 - (-1)^5 - 1 \cdot i \\
 &= i + -1 + 1 - i \\
 &= 0_{\#}
 \end{aligned}$$

$$f) i^{-1} + i^{-2} - i^{-3} - i^{-4}$$

$$= \frac{1}{i} + \frac{1}{i^2} - \frac{1}{i^3} - i^{4 \times -1}$$

$$= \frac{i^4}{i} + \frac{i^4}{i^2} - \frac{i^4}{i^3} - 1 \quad [\because i^4 = 1]$$

$$= i^3 + i^2 - i - 1$$

$$= i^2 \cdot i - 1 - i - 1$$

$$= -i - 1 - i - 1$$

$$= -2 - 2i$$

12. Prove that:

$$a. i^n + i^{n+1} + i^{n+2} + i^{n+3} = 0$$

→ L.H.S,

$$i^n + i^n \cdot i + i^n \cdot i^2 + i^n \cdot i^3$$

$$= i^n (1 + i + i^2 + i^2 \cdot i)$$

$$= i^n (1 + i - i - i)$$

$$= i^n \times 0$$

$$= 0 = R.H.S$$

proved

$$b) (1+i)^4 \left(1 + \frac{1}{i}\right)^4 = 16$$

→ L.H.S,

$$((1+i)^2)^2 \left(1 + \frac{i^4}{i}\right)^4 \quad [\because i^4 = 1]$$

$$= (1 + 2i + i^2)^2 (1 + i^3)^4$$

$$= (1 + 2i - 1)^2 (1 + i^3)^4$$

$$= 4i^2 (1 + i^3)^4$$

$$\begin{aligned}
 &= -4(1+i^2 \cdot i)^4 \\
 &= -4[(1-i)^2]^4 \\
 &= -4[(1-2i+i^2)]^2 \\
 &= -4(1-2i-1)^2 \\
 &= -4(-2i)^2 \\
 &= -4 \times 4i^2 \\
 &= -4 \times -4 \\
 &= 16 = R.H.S
 \end{aligned}$$

Proved.

13.

a) If $\sqrt{x-iy} = a-ib$, prove that $\sqrt{x+iy} = a+ib$.

→ Soln,

$$\sqrt{x-iy} = a-ib$$

Squaring both side,

$$x-iy = (a-ib)^2$$

$$x-iy = a^2 - 2 \cdot a \cdot ib + b^2 i^2$$

$$x-iy = a^2 - b^2 - i \cdot 2ab$$

equating real and imaginary parts on both side,

$$x = a^2 - b^2$$

$$y = 2ab$$

Now,

$$\begin{aligned}
 \sqrt{x+iy} &= \sqrt{a^2 - b^2 + 2ab \cdot i} \\
 &= \sqrt{a^2 + b^2(i^2) + 2ab \cdot i} \\
 &= \sqrt{(a^2 + 2 \cdot a \cdot ib + (ib)^2)} \\
 &= \sqrt{(a+ib)^2} \\
 &= a+ib
 \end{aligned}$$

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b) If $\sqrt{a+ib} = 2-3i$, prove that $\sqrt{a-ib} = 2+3i$

→ Soln,

$$\sqrt{a+ib} = 2-3i$$

Squaring both side,

$$a+ib = (2-3i)^2$$

$$a+ib = 4 - 2 \cdot 2 \cdot 3i + 9i^2$$

$$a+ib = 4 - 12i - 9$$

$$a+ib = -5 - 12i$$

equating real and imaginary parts
on both side,

$$a = -5$$

$$b = -12$$

Now,

$$\begin{aligned}\sqrt{a-ib} &= \sqrt{-5 - (-12i)} \\&= \sqrt{-5 + 12i} \\&= \sqrt{4 + 12i + 9i^2} \quad [\because i^2 = -1] \\&= \sqrt{2^2 + 2 \cdot 2 \cdot 3i + (3i)^2} \\&= \sqrt{(2+3i)^2} \\&= 2+3i \\&\quad \#.\end{aligned}$$

Qn If $\sqrt{a-ib} = 4-5i$ then prove that $\sqrt{a+ib} = 4+5i$

→ Soln,

$$\sqrt{a-ib} = 4-5i$$

Squaring both side,

$$a-ib = (4-5i)^2$$

$$a-ib = (4)^2 - 2 \cdot 4 \cdot 5i + (5i)^2$$

$$a-ib = 16 - 40i + 25i^2$$

$$a-ib = 16 - 40i - 25 \quad [\because i^2 = -1]$$

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$$a - ib = -9 - 40i$$

equating L.H.S and R.H.S,

$$a = -9$$

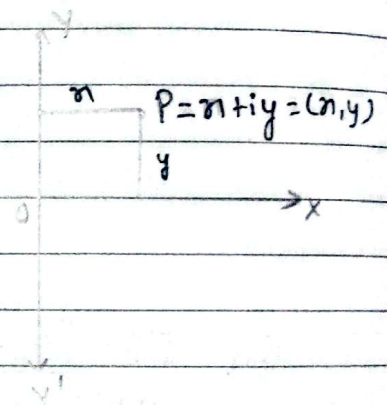
$$b = 40$$

Now,

$$\begin{aligned}\sqrt{a+ib} &= \sqrt{-9+40i} \\ &= \sqrt{-25+16+40i} \\ &= \sqrt{25i^2 + 2 \cdot 5i \cdot 4 + 4^2} \quad [\because i^2 = -1] \\ &= \sqrt{(5i)^2 + 2 \cdot 5i \cdot 4 + 4^2} \\ &= \sqrt{(4+5i)^2} \\ &= 4+5i \\ &\quad \text{Proved}\end{aligned}$$

• Geometrical Representation of Complex Number:

→ let xox' and yoy' be two mutually perpendicular straight lines respectively called real axis and Imaginary axis.



A complex number $Z = x + iy$ is represented as a point $P(x, y)$ in complex plane where,

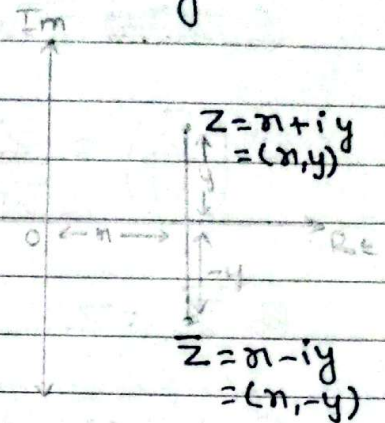
x = Real Part of Z = length of perpendicular
 y = Imaginary part of Z = P to M .

Conjugate of complex Number

→ Conjugate of a complex number $z = x + iy$ is denoted by $\bar{z}$ and is defined by $\bar{z} = \overline{x + iy} = x - iy$.

For e.g.:

- $\overline{2 - 3i} = 2 + 3i$
- $\overline{-4 + 8i} = -4 - 8i$
- $\overline{3i + 4} = -3i + 4$
- $\overline{4i} = -4i$
- $\overline{5} = 5$



Geometrically, complex conjugate represents the reflection of given complex number about real axis.

Notes

① A complex number z is purely real iff $\bar{z} = z$.

For eg: If $z = 2$ then $\bar{z} = 2 = z$
i.e. $\bar{z} = z$

② A complex number z is purely imaginary iff $\bar{z} = -z$.

For eg: If $z = 4i$ then $\bar{z} = -4i = -z$
i.e. $\bar{z} = -z$.

• Properties of Complex conjugate:

Let, $z = a + ib$ and $w = c + id$ be two complex

numbers, then

$$① \overline{Z+W} = \overline{Z} + \overline{W}$$

$$② \overline{Z \cdot W} = \overline{Z} \cdot \overline{W}$$

$$③ \overline{\left(\frac{Z}{W}\right)} = \frac{\overline{Z}}{\overline{W}} ; W \neq 0$$

$$④ \overline{\overline{Z}} = Z$$

$$⑤ \operatorname{Re}(Z) = \frac{1}{2} (Z + \overline{Z}) \quad \text{ie. } Z + \overline{Z} = 2 \operatorname{Re}(Z)$$

Proof

Let, $Z = a + ib$ be any complex number,
then,

$$\operatorname{Re}(Z) = a$$

$$\text{and } \overline{Z} = a - ib$$

Now,

$$\begin{aligned} Z + \overline{Z} &= a + ib + a - ib \\ &= 2a \end{aligned}$$

$$\therefore Z + \overline{Z} = 2 \operatorname{Re}(Z)$$

$$⑥ \operatorname{Im}(Z) = \frac{1}{2i} (Z - \overline{Z}) \quad \text{ie. } Z - \overline{Z} = 2i \operatorname{Im}(Z)$$

Proof

Let, $Z = a + ib$ any complex number,
then,

$$\operatorname{Im}(Z) = b$$

$$\text{and, } \overline{Z} = a - ib$$

Now,

$$\begin{aligned} z - \bar{z} &= a+ib - (a-ib) \\ &= a+ib - a + ib \\ &= 2ib \\ &= 2i \operatorname{Im}(z) \end{aligned}$$

$$\therefore z - \bar{z} = 2i \operatorname{Im}(z) \quad \#.$$

$$\textcircled{7} \quad \overline{(z^2)} = (\bar{z})^2$$

Q.7 Find the Conjugate of following complex number.

$$\begin{aligned} \text{a) } (1+2i)^2 &= 1 + 2 \cdot 1 \cdot 2i + (2i)^2 \\ &= 1 + 4i + 4i^2 \\ &= 1 + 4i - 4 \quad [\because i^2 = -1] \\ &= -3 + 4i \end{aligned}$$

Now,

$$\text{Conjugate of } (1+2i)^2 = -3 - 4i \quad \#.$$

$$\begin{aligned} \text{b) } (1+4i)(3-5i) &= 3 - 5i + 12i - 20i^2 \\ &= 3 + 7i + 20 \quad [i^2 = -1] \\ &= 23 + 7i \end{aligned}$$

Now,

$$\text{Conjugate of } (1+4i)(3-5i) = 23 - 7i \quad \#.$$

$$\begin{aligned} \text{c) } \frac{(6,8)}{(1,-2)} &= \frac{6+8i}{1-2i} \end{aligned}$$

$$\begin{aligned}
&= \frac{6+8i}{1-2i} \times \frac{1+2i}{1+2i} \\
&= \frac{6+12i+8i+16i^2}{1-(2i)^2} \\
&= \frac{6+20i-16}{1-4i^2} \quad [\because i^2 = -1] \\
&= \frac{-10+20i}{1+4} \\
&= \frac{-10+20i}{5} \\
&= -\frac{10}{5} + \frac{20}{5}i \\
&= -2+4i
\end{aligned}$$

Now,

$$\text{Conjugate of } \frac{(6,8)}{(1,-2)} = -2-4i \quad \#.$$

Absolute value or modulus of Complex Number:

↳ The absolute value or modulus of a complex number $Z=a+ib$ is denoted by $|Z|$ and is defined by $|Z| = \sqrt{a^2+b^2}$.

For e.g: $z = 2-3i$

$$\begin{aligned}
|Z| &= \sqrt{2^2 + (-3)^2} \\
&= \sqrt{4+9} \\
&= \sqrt{13} \quad \#.
\end{aligned}$$

Geometrically, absolute value of a complex number represents the distance of that complex number from origin in the complex plane.

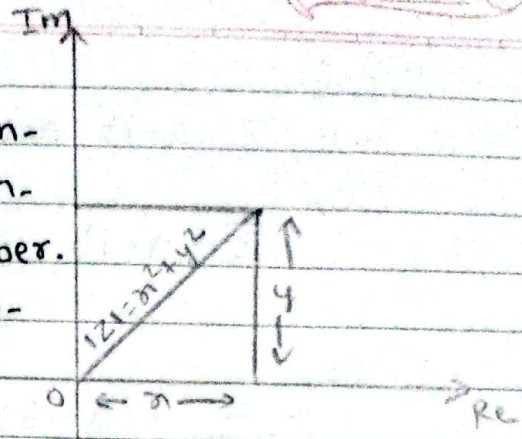


Fig - absolute value of Complex Number

Note:

→ The absolute value of a complex number z is a non-negative real number.
i.e. $|z| \geq 0$.

Properties of absolute value or modulus of complex number:

For any two complex numbers z and w ,

$$① |z| = |\bar{z}| = |-z| = |-\bar{z}|$$

$$② |zw| = |z||w|$$

$$③ \left| \frac{z}{w} \right| = \frac{|z|}{|w|} ; w \neq 0$$

$$④ z \cdot \bar{z} = |z|^2$$

Proof

let $z = a + ib$ be any complex number.

So,

$$\bar{z} = a - ib \text{ and } |z| = \sqrt{a^2 + b^2}$$

L.H.S,

$$z \cdot \bar{z} = (a + ib)(a - ib)$$

$$= a^2 - i^2 b^2$$

$$= a^2 + b^2 \quad [\because i^2 = -1]$$

$$= (\sqrt{a^2 + b^2})^2$$

$$\therefore z \cdot \bar{z} = |z|^2$$

$$\textcircled{5} |z^2| = |z|^2$$

$$\textcircled{6} |z| = 0 \text{ iff } z = 0$$

$$\textcircled{7} -|z| \leq \operatorname{Re}(z) \leq |\operatorname{Re}(z)| \leq |z|$$

and,

$$-|z| \leq \operatorname{Im}(z) \leq |\operatorname{Im}(z)| \leq |z|$$

Qn Find the absolute value of followings:

$$a) (1 - 2i)^2$$

$$\rightarrow \text{let } z = (1 - 2i)^2 \quad |z| = |1 - 2i|^2$$

$$= (\sqrt{1^2 + (-2)^2})^2$$

$$= (\sqrt{1 + 4})^2$$

$$= 1 + 4$$

$$= 5 \text{ Ans.}$$

b) $(2+3i)(4-3i)$

$$\begin{aligned} \text{Absolute Value} &= |2+3i| \cdot |4-3i| \\ &= \sqrt{2^2+3^2} \cdot \sqrt{4^2+(-3)^2} \\ &= \sqrt{4+9} \cdot \sqrt{16+9} \\ &= \sqrt{13} \cdot \sqrt{25} \\ &= 5\sqrt{13} \text{ u.} \end{aligned}$$

c) $\frac{(6,8)}{(1,-2)}$

→ Soln,

$$\begin{aligned} \text{Absolute Value} &= \frac{|(6,8)|}{|(1,-2)|} \\ &= \frac{\sqrt{6^2+8^2}}{\sqrt{1^2+(-2)^2}} \\ &= \frac{\sqrt{36+64}}{\sqrt{1+4}} \\ &= \frac{\sqrt{100}}{\sqrt{5}} \\ &= \frac{10}{\sqrt{5}} \\ &= 2\sqrt{5} \text{ u.} \end{aligned}$$

Sumit
05/03

Exercise 7.2

1. Find the Conjugate of

a) $2+6i$

→ Conjugate = $2-6i$

b) $-2i+1$

→ Conjugate = $2i+1$
 $= 1+2i$

c) $(2-3i)(4+7i)$

→ $8+14i-12i-21i^2$

$= 8+2i+21$ ($\because i^2 = -1$)

$= 29+2i$

Conjugate = $29-2i$

d) $(1,3), (-2,5)$

→ $(1+3i)(-2+5i)$

$= -2+5i-6i+15i^2$

$= -2-i-15$ ($\because i^2 = -1$)

$= -17-i$

Conjugate = $-17+i$

$= (-17,1)$ #.

e) $(2,-4)$

$(-1,1)$

→ $(2-4i)$

$(-1+i)$

$= \frac{(2-4i)}{(-1+i)} \times \frac{(-1-i)}{(-1-i)}$

$= \frac{-2-2i+4i+4i^2}{(-1)^2 - i^2}$

$= \frac{-2+2i-4}{1+1}$ ($\because i^2 = -1$)

$= \frac{-6+2i}{2}$

$= -3+i$

Conjugate = $-3-i$

$= (-3,-1)$ #.

f) $3+4i$

$3-4i$

→ $\frac{3+4i}{3-4i} \times \frac{3+4i}{3+4i}$

$= \frac{(3+4i)^2}{3^2 - (4i)^2}$

$= \frac{3^2 + 2 \cdot 3 \cdot 4i + (4i)^2}{9 - 16i^2}$

$= \frac{9 + 24i - 16}{9 + 16}$ ($\because i^2 = -1$)

$= \frac{-7+24i}{25}$

$= -\frac{7}{25} + \frac{24}{25}i$

Conjugate = $-\frac{7}{25} - \frac{24}{25}i$ #.

$$\begin{aligned}
 g) (2-i)^{-1} &= \frac{1}{(2-i)} \\
 &= \frac{1}{(2-i)} \times \frac{(2+i)}{(2+i)} \\
 &= \frac{2+i}{2^2 - i^2} \\
 &= \frac{2+i}{4+1} \quad (\because i^2 = -1) \\
 &= \frac{2}{5} + \frac{i}{5}
 \end{aligned}$$

$$\text{Conjugate} = \frac{2}{5} - \frac{1}{5}i$$

$$\begin{aligned}
 h) (3+4i)^2 &= 3^2 + 2 \cdot 3 \cdot 4i + (4i)^2 \\
 &= 9 + 24i + 16i^2 \\
 &= 9 + 24i - 16 \quad (\because i^2 = -1) \\
 &= -7 + 24i \\
 \text{conjugate} &= -7 - 24i
 \end{aligned}$$

2. If $Z_1 = (3, 2)$, $Z_2 = (5, 3)$, find

a) $\overline{Z_1 + Z_2}$

$$\begin{aligned}
 \rightarrow Z_1 + Z_2 &= (3, 2) + (5, 3) \\
 &= (3+5, 2+3) \\
 &= (8, 5) \\
 &= 8+5i
 \end{aligned}$$

Now,

$$\begin{aligned}
 \text{Conjugate of } Z_1 + Z_2 &= \overline{Z_1 + Z_2} = 8 - 5i \\
 &= (8, -5)
 \end{aligned}$$

b) $\overline{Z_1 - Z_2}$

$$\begin{aligned}
 \rightarrow Z_1 - Z_2 &= (3, 2) - (5, 3) \\
 &= (3-5, 2-3) \\
 &= (-2, -1) = -2-i
 \end{aligned}$$

Now,

$$\overline{Z_1 - Z_2} = (-2, 1) \text{ or } -2+i$$

$$c) \overline{\left(\frac{Z_1}{Z_2}\right)}$$

$$= \frac{\overline{Z_1}}{\overline{Z_2}}$$

$$= \frac{(3, 2)}{(5, 3)}$$

$$= \frac{3+2i}{5+3i}$$

$$= \frac{3+2i}{5+3i} \times \frac{5-3i}{5-3i}$$

$$= \frac{15+10i-9i-6i^2}{5^2-(3i)^2}$$

$$= \frac{15+i+6}{25+9} \quad (\because i^2 = -1)$$

$$= \frac{21+i}{34}$$

$$= \frac{21}{34} + \frac{1}{34}i$$

Now,

$$\overline{\left(\frac{Z_1}{Z_2}\right)} = \frac{21}{34} - \frac{1}{34}i \quad \text{or} \quad \left(\frac{21}{34}, -\frac{1}{34}\right)$$

$$d) \overline{Z_1 \cdot Z_2}$$

$$= \overline{(3, 2) \cdot (5, 3)}$$

$$\rightarrow Z_1 \cdot Z_2 = (3, 2) \cdot (5, 3)$$

$$= (3+2i)(5+3i)$$

$$= 15+10i+9i+6i^2$$

$$= 15+19i-6 \quad (\because i^2 = -1)$$

$$= 9+19i$$

Now,

$$\overline{Z_1 \cdot Z_2} = 9-19i \quad \text{or} \quad (9, -19)$$

3. Find the absolute value of

a) $2+3i$

$$\begin{aligned} \rightarrow \text{let, } Z &= 2+3i \\ |Z| &= \sqrt{2^2+3^2} \\ &= \sqrt{4+9} \\ &= \sqrt{13} \text{ u.} \end{aligned}$$

b) $2-5\sqrt{3}i$

$$\begin{aligned} \rightarrow \text{let, } Z &= 2-5\sqrt{3}i \\ |Z| &= \sqrt{2^2+(-5\sqrt{3})^2} \\ &= \sqrt{4+75} \\ &= \sqrt{79} \text{ u.} \end{aligned}$$

c) $6i-1$

$= -1+6i$

$$\begin{aligned} \text{let, } Z &= -1+6i \\ |Z| &= \sqrt{(-1)^2+6^2} \\ &= \sqrt{1+36} \\ &= \sqrt{37} \text{ u.} \end{aligned}$$

d) $(3+4i)(4+i)$

$\rightarrow \text{let } Z = (3+4i)$ ~~$(4+i)$~~

and, $X = (4+i)$

Now,

$$\begin{aligned} |Z| \cdot |X| &= |3+4i| \cdot |4+i| \\ &= \sqrt{3^2+4^2} \cdot \sqrt{4^2+1^2} \\ &= \sqrt{9+16} \cdot \sqrt{16+1} \\ &= \sqrt{25} \cdot \sqrt{17} \\ &= 5\sqrt{17} \text{ u.} \end{aligned}$$

e) $\frac{4+5i}{4-5i}$

$\rightarrow \text{let, } X = 4+5i$
and, $Z = 4-5i$

Now,

$$\begin{aligned} \left| \frac{X}{Z} \right| &= \frac{|X|}{|Z|} \\ &= \frac{|4+5i|}{|4-5i|} \\ &= \frac{\sqrt{4^2+5^2}}{\sqrt{4^2+(-5)^2}} \\ &= \frac{\sqrt{16+25}}{\sqrt{16+25}} \\ &= 1 \text{ u.} \end{aligned}$$

f) $\frac{1-i}{2+5i}$

$\rightarrow \text{let, } X = (1-i)$ and,
 $Z = (2+5i)$

Now,

$$\begin{aligned} \left| \frac{X}{Z} \right| &= \frac{|X|}{|Z|} \\ &= \frac{|1-i|}{|2+5i|} \\ &= \frac{\sqrt{1^2+(-1)^2}}{\sqrt{2^2+5^2}} \\ &= \frac{\sqrt{1+1}}{\sqrt{4+25}} \\ &= \frac{\sqrt{2}}{\sqrt{29}} \text{ u.} \end{aligned}$$

g) $(4-3i)^{-1}$
 $\rightarrow \text{let, } Z = (4-3i)^{-1}$

$$= \frac{1}{4-3i}$$

$$|Z| = \frac{|1|}{|4-3i|}$$

$$= \frac{\sqrt{1^2+0^2}}{\sqrt{4^2+(-3)^2}}$$

$$= \frac{\sqrt{1}}{\sqrt{16+9}}$$

$$= \frac{1}{\sqrt{25}}$$

$$= \frac{1}{5}$$

4. If $Z=2+3i$ and $W=1-2i$, verify that

a) $|Z \cdot W| = |Z| \cdot |W|$

$\rightarrow$ L.H.S,

$$\begin{aligned} |Z \cdot W| &= |(2+3i)(1-2i)| \\ &= |2-4i+3i-6i^2| \\ &= |2-i+6| \quad (\because i^2 = -1) \\ &= |8-i| \\ &= \sqrt{8^2+(-1)^2} \\ &= \sqrt{64+1} \\ &= \sqrt{65} \end{aligned}$$

R.H.S = $|Z| \cdot |W|$

$$\begin{aligned} &= |2+3i| \cdot |1-2i| \\ &= \sqrt{2^2+3^2} \cdot \sqrt{1^2+(-2)^2} \\ &= \sqrt{4+9} \cdot \sqrt{1+4} \\ &= \sqrt{13} \cdot \sqrt{5} \end{aligned}$$

$$= \sqrt{13 \times 5}$$
$$= \sqrt{65}$$

$$\therefore \text{L.H.S} = \text{R.H.S} = \sqrt{65} \quad \checkmark$$

$$b) \left| \frac{Z}{W} \right| = \frac{|Z|}{|W|}$$

$\rightarrow$ L.H.S,

$$\left| \frac{Z}{W} \right| = \left| \frac{2+3i}{1-2i} \right|$$

$$= \left| \frac{2+3i}{1-2i} \times \frac{1+2i}{1+2i} \right|$$

$$= \left| \frac{2+4i+3i+6i^2}{1^2-4i^2} \right|$$

$$= \left| \frac{2+7i-6}{1+4} \right| \quad (\because i^2 = -1)$$

$$= \left| \frac{-4+7i}{5} \right|$$

$$= \left| -\frac{4}{5} + \frac{7}{5}i \right|$$

$$= \sqrt{\left(-\frac{4}{5}\right)^2 + \left(\frac{7}{5}\right)^2}$$

$$= \sqrt{\frac{16}{25} + \frac{49}{25}}$$

$$= \sqrt{\frac{65}{25}}$$

$$= \sqrt{\frac{13}{5}}$$

Now,

$$\text{R.H.S} = \frac{|Z|}{|W|}$$

$$\begin{aligned}
 &= \frac{|2+3i|}{|1-2i|} \\
 &= \frac{\sqrt{2^2+3^2}}{\sqrt{1^2+(-2)^2}} \\
 &= \frac{\sqrt{4+9}}{\sqrt{1+4}} \\
 &= \frac{\sqrt{13}}{\sqrt{5}} \\
 &= \sqrt{\frac{13}{5}}
 \end{aligned}$$

$$\therefore \text{L.H.S} = \text{R.H.S} = \sqrt{\frac{13}{5}} \quad \text{H}$$

$$\text{C. } |Z+W| \leq |Z| + |W|$$

→ L.H.S,

$$\begin{aligned}
 &|2+3i+1-2i| \\
 &= |3+i| \\
 &= \sqrt{3^2+1^2} \\
 &= \sqrt{9+1} \\
 &= \sqrt{10} = 3.16
 \end{aligned}$$

R.H.S,

$$\begin{aligned}
 &|Z| + |W| \\
 &= |2+3i| + |1-2i| \\
 &= \sqrt{2^2+3^2} + \sqrt{1^2+(-2)^2} \\
 &= \sqrt{4+9} + \sqrt{1+4} \\
 &= \sqrt{13} + \sqrt{5} = 5.84
 \end{aligned}$$

$$\therefore |Z+W| \leq |Z| + |W|$$

Proved